

Lab 4 Exercises

Assignment No1.

Referencing useHover, design a custom hook useClickPosition that takes a logName as input. When the user clicks within a designated area, it should output a log with the click details and indicate which area was clicked.

Some hints:

(1) You can design it based on the structure shown in the following example.

```
const useClickPosition = ({logName = ''}) => {
  const [clickPosition, setClickPosition] = useState({ x: 0, y: 0 });
  const elementRef = useRef(null);

  useEffect(() => {
    // ...
  });

  if (element) {
    // ...
  }

  return () => {
    // ...
  };
}, [logName]);

return [elementRef, clickPosition];
};
```

(2) When a component needs to attach multiple refs, you can use the following approach:

```
const setRefs = (element) => {
  hubHoverRef.current = element;
  HubclickPositionRef.current = element;
};

return (
  <div ref={setRefs} style={styles.hub}>
```

Finally, it can show the logs like this:

```
hub clickPosition:  ▶ {x: 566, y: 235.0703125}
hub clickPosition:  ▶ {x: 392, y: 231.0703125}
hub clickPosition:  ▶ {x: 273, y: 224.0703125}
```

Assignment No 2.

Refactor the dark mode code from Lab 3 using the context hook.

Assignment No 3.

Referring to the example of LatestNews in the router, set up the gallery designed in Lab 3 as a new page with the URL `"/gallery"`.

